Ariel Lubonja

PhD Candidate, Computer Science: **Distributed Systems & Parallel Computing**. **Graduating 05/28**
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Education

Johns Hopkins University

PhD, Computer Science — GPA: 3.92/4.0 — Advisors: R. Burns, J. Vogelstein, A. Yuille — *Aug 2022 – May 2028*

MS, Data Science — GPA: 3.8/4.0 — Dean's List — *Jan 2021 – Aug 2022*

Research Experience

Google Research Collaboration — Optimizing Google YDF — May 2025 – Present

- Delivered 8 PRs reducing Google YDF's oblique training time 6x (12 h to 2 h) in collaboration w/ Google Zurich.
- Designed two novel kernels (1.7–2.5x) using **SIMD** (AVX-2, AVX-512) vectorized binning and dynamic splitting. Devised a memory-efficient approach by analyzing memory access patterns.
- Mentored junior PhD student in implementing this novel approach in **CUDA**, achieving 25x speedup over CPU.
- Sped up high-dimensional gradient-boosted trees (GBTs) 50x by **perf profiling** and fixing a mutex bottleneck.
- Designing the first sparse oblique **distributed computing** GBT, which is a memory and communication problem.

Scalable Node Embedding for Graph ML — Dec 2023 – May 2024

- Cut Graph Encoder Embedding (GEE node embedding) time of billion-edge graphs from 56 min to 6.4 s on 24 cores (500x vs. Python, 17x vs. Numba JIT) via a lock-free, edge-parallel, memory-bandwidth-bound Ligra rewrite.
- Open-source package and IPDPS-W 2024 (GrAPL) paper.

NVIDIA PhysicsNeMo Contributor — Nov 2025 – Feb 2026

- Implemented and tested a new API in NVIDIA PhysicsNeMo (PyTorch); merged upstream in 2 pull requests.
- Improved runtime by 2.5x on low-resolution inputs by eliminating a redundant API call to Johns Hopkins Turbulence Database, working together with a NVIDIA engineer.

LLM Inference Efficiency & Fine-Tuning — Mar 2024 – Jul 2024

- Coded a low-latency GPU early-exit of GPT-2 by studying agreement between intermediate and final outputs.
- Applied LoRA fine-tuning of BLOOM 560M–3B (0.28% trainable params, fp16, 8-bit inference) on T4/A100 GPUs to pharmaceutical-private data, helping a former colleague.

GPU-Accelerated Orthogonal Matching Pursuit - Signal Processing — May 2022 – Jan 2023

- Built a 310x faster drop-in GPU replacement for scikit-learn's Orthogonal Matching Pursuit (OMP), using batched **cuBLAS** fast linear algebra library (in PyTorch), with Sebastian Praesius, PhD.
- Published as Python package batched-omp, arXiv preprint and open-source code.

Work Experience

Software Developer — May 2017 – Jan 2021

jBoxers — *Sofia, Bulgaria*

- NLP chatbot for USC (Mycroft, Dialogflow); Library of Congress Virtual Card Catalog (Angular frontend, Django and Java EE backend).
- Led a small Angular UI project with 2 summer interns.

Skills

Languages & Tools: **C/C++**, Python, CUDA, Java, Linux, Git, Bazel, LLVM, gcc, Hadoop, **OpenMP**
Performance Engineering: **Distributed Clusters, Parallel Programming**, **Intel VTune, perf**, **NVIDIA Nsight Systems**, **CPU/GPU kernel optimization**, **memory-hierarchy**, **SIMD**, HPC, cuBLAS, Zarr Distributed Storage
Machine Learning: PyTorch, Tensorflow, **Graphs**, Vertex AI, Tree Ensembles, Computer Vision, , XLA

Leadership & Honors

Teaching Assistant: **Big Data Systems** (current) — Ray, Spark, Kafka, Hadoop, NoSQL
Parallel Programming & Performance Engineering, (2022–2024) — SIMD, multicore, memory, CUDA
Awards: Fulbright-style fully funded Albanian fellowship. 1 of 14 nationwide
Leadership: Vice President, Upsilon Pi Epsilon, Department representative at Graduate Representative Organization